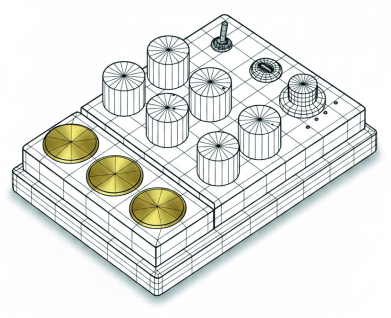
// generative instrument

SixOp_OS

diceFriend / quick start



The open-source diceFriend showcase: six-operator FM, three voices and eight algorithms.

START: USB power + stereo audio out; keep listening level low. Starts in AS: a three-voice drone. TS: three played voices.

// KNOBS / AS

P	NORMAL	SHIFT
1	Pitch: C1-C6	LFO depth
2	Chord: eight voicings	LFO rate
3	Algorithm: eight routings	Detune
4	Ratio: harmonic to metallic	Pulse: left = pure drone
5	Depth: FM index	Grit: lo-fi colour
6	Feedback	Space: stereo echo

// KNOBS / TS

P	NORMAL	SHIFT
1	Voice 1 pitch	Algorithm
2	Voice 2 pitch	Ratio
3	Voice 3 pitch	Depth
4	Voice 1 decay	Feedback
5	Voice 2 decay	Contour
6	Voice 3 decay	Detune

Pickup: move 8% from the armed position; takeover glides for 200 ms. AS Grit, Space and LFO Rate remain active in TS.

// PADS / DICE

AS pads: T1 Bright / T2 Metal / T3 Growl; touches re-strike. **TS:** T1-T3 play voices; hold for gate. Tine/Bell decay; Bass/Brass sustain.

NS tap: AS randomize NS; TS next Tine/Bell/Bass/Brass. **SH tap <1 s:** randomize SH. **NS hold 3 s:** AS/TS. **SH hold 10 s:** bootloader.

// CV / CLOCK

CV1: priority trigger. CV2: pulse clock or steady FM depth. CV > MIDI > internal: MIDI after 20 s without CV, internal after 30 s without external activity.

// USB MIDI

IN ch 1-15: notes, release, Clock 24 PPQN, transport; CC20-31 panel. OUT ch 16: local notes, panel CC, CV2 CC74; CV-derived Clock.

OPEN SOURCE: the complete commented reference implementation, renderer, developer notes and MIT license are available in the SixOp_OS GitHub package.